

AI Agent Research Compendium — Volume 4: AI Agent Security

The most critical security papers for production AI agent deployment. Covers attacks, defenses, protocol security, and governance.

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1. Attack Vectors & Vulnerabilities

Internal Safety Collapse in Frontier LLMs (arXiv: 2603.23509)

Agents produce harmful content as side effect of normal tasks — 56 cross-domain scenarios. **Critical awareness paper.**

Malicious Agent Skills in the Wild (arXiv: 2602.06547)

98K agent skills analyzed from community registries for malicious plugins. **Must-read for marketplace security.**

Beyond Max Tokens: Stealthy Resource Amplification (arXiv: 2601.10955)

Multi-turn economic DoS exploiting agent-tool loop — costs inflated up to 658x via MCP modifications. **Critical for cost management.**

Persuasion Propagation in LLM Agents (arXiv: 2602.00851)

User persuasion during conversation carries over to later autonomous agent tasks — novel social engineering vector.

Mandela Effect in LLM Multi-Agent Systems (arXiv: 2602.00428)

Collective false memories form in MAS with defenses including cognitive anchoring.

2. Prompt Injection & Defense

Prompt Injection Attacks on Agentic Coding Assistants (arXiv: 2601.17548)

Survey of 78 studies systematizing attacks with three-dimensional taxonomy. **Definitive reference.**

Learning to Inject: Automated Prompt Injection via RL (arXiv: 2602.05746)

Uses RL to auto-generate injection attacks that transfer across frontier models.

VIGIL: Defending Against Tool Stream Injection (arXiv: 2601.05755)

Verify-before-commit protocol with speculative hypothesis generation and intent-grounded verification.

StepShield: When to Intervene on Rogue Agents (arXiv: 2601.22136)

Benchmark for when violations are detected during execution, with temporal metrics for early intervention.

Too Helpful to Be Safe: User-Mediated Attacks (arXiv: 2601.10758)

Commercial agents handle adversarial user instructions without explicit safety requests.

3. Protocol Security (MCP, A2A)

Breaking the Protocol: MCP Security Analysis (arXiv: 2601.17549)

First security analysis of MCP specification — three protocol-level vulnerabilities with backward-compatible fixes. **Must-read for MCP.**

SMCP: Secure Model Context Protocol (arXiv: 2602.01129)

Unified identity management, mutual authentication, and fine-grained policy enforcement for MCP.

MCP-ITP: Implicit Tool Poisoning in MCP (arXiv: 2601.07395)

Poisoned tool metadata manipulates agent via legitimate tools without invocation. **Novel attack surface.**

MCP-SandboxScan: WASM-based MCP Tool Execution (arXiv: 2601.01241)

WebAssembly sandbox for untrusted MCP tools with auditable reports.

Agent2Agent Threats in Safety-Critical LLMs (arXiv: 2602.05877)

Attack paths in A2A protocols for automotive assistants, from distraction to unauthorized control.

4. Agent Governance & Access Control

Faramesh: Protocol-Agnostic Execution Control Plane (arXiv: 2601.17744)

Authorization boundaries with canonical action representation and deterministic policy evaluation. **Essential for enterprise governance.**

Mandatory Access Control for LLM Agent Systems (arXiv: 2601.11893)

Information flow graph monitoring with attribute-based policies against privilege escalation.

AgentTRIM: Tool Risk Mitigation (arXiv: 2601.12449)

Offline interface verification and runtime least-privilege tool access with adaptive filtering.

AgentGuardian: Learning Access Control Policies (arXiv: 2601.10440)

Context-aware access-control policies learned from monitored execution traces.

CaMeLs: System-level Security for Computer Use Agents (arXiv: 2601.09923)

Provable control flow integrity against prompt injection while preserving capability.

Delegation Without Living Governance (arXiv: 2601.21226)

Runtime governance to preserve human relevance in agent-driven decision-making at machine speed.

GRACE: Neuro-Symbolic Architecture for Safe Alignment (arXiv: 2601.10520)

Decoupling normative reasoning from instrumental decision-making via Moral Module and compliance Guard.

Agent Governance in Healthcare (arXiv: 2601.15630)

Five control-plane layers for agent fleet governance: identity registry, orchestration, policy enforcement.

5. Privacy & Data Protection

SoK: Privacy Risks and Mitigations in RAG (arXiv: 2601.03979)

Systematized privacy risks, mitigations, and evaluation strategies. **Reference paper.**

NeuroFilter: Privacy Guardrails for Conversational Agents (arXiv: 2601.14660)

Activation-space guardrails detecting privacy-violating intent via internal representation separation.

Memory Poisoning Attack and Defense (arXiv: 2601.05504)

Input/output moderation with trust scoring and memory sanitization with trust-aware retrieval.

Making Theft Useless: Adulteration-Based KG Protection (arXiv: 2601.00274)

Pre-emptively injecting false entries to make stolen GraphRAG KGs unusable. **Novel IP protection.**

Agent Identity URI Scheme (arXiv: 2601.14567)

agent:// URI scheme with trust roots, hierarchical capability paths, and cryptographic attestation.

30+ security papers. Selected for practical attacks with demonstrated exploits, MCP/A2A protocol security, and production governance frameworks.